



(EN) Vitamin D rapid test cassette

Package leaflet For self-testing REF OVD-402H

A rapid test for the semi-quantitative detection of 25-hydroxyvitamin D in human fingerstick blood. For in vitro diagnostic use for self-testing.

PURPOSE

The Vitamin D Rapid Test Cartridge is a rapid chromatographic immunoassay for the semi-quantitative detection of 25-hydroxyvitamin D (25(OH)D) in human whole fingerstick blood. This assay provides a preliminary diagnostic test result and can be used to screen for vitamin D deficiency.

SUMMARY

Vitamin D belongs to a group of fat-soluble secosteroids responsible for increasing the intestinal absorption of calcium, iron, magnesium, phosphate and zinc. In humans, the most important compounds in this group are vitamin D3 and vitamin D2. Vitamin D3 is produced naturally in human skin by exposure to ultraviolet light, and vitamin D2 is mainly obtained from foods.

Vitamin D is transported to the liver where it is metabolized to 25-hydroxyvitamin D. A blood test for 25-hydroxyvitamin D is used in meth to determine the concentration of vitamin D in the body. The blood concentration of 25-hydroxyvitamin D (including D2 and D3) A is considered the best indicator of vitamin D status. Vitamin D deficiency is now recognized as a global epidemic. Virtually every cell in our body has receptors for vitamin D, which means they all need a "sufficient" level of vitamin D to function adequately. The health risks associated with vitamin D deficiency are much more serious than previously thought. Vitamin deficiency has been linked to a variety of serious diseases: osteoporosis, osteomalacia, multiple sclerosis, cardiovascular disease, pregnancy complications, diabetes, depression, strokes, autoimmune diseases, influenza. Various cancers, infectious diseases, Alzheimer's, obesity and high mortality, etc.

PRINCIPLE

The vitamin D test is an immunoassay based on the principle of competitive binding. In testing, the mixture migrates up the membrane chromatographically by capillary action. The membrane is

pre-coated with 25(OH)D antigens on the test line area of the strip. During testing, the 15(OH)D present in the sample will compete with the 25(OH)Don on the test line for a limited amount of anti-25 OH vitamin D antibody in the conjugate. The higher the concentration of 25(OH)D in the sample, the lighter the T line will be. The result will be read according to the color card provided with the kit. To serve as a procedural control, a colored line will always appear in the area of the control line indicating that an appropriate volume of sample has been added and membrane firing has occurred.

PRECAUTIONS

Please read the information in this leaflet before carrying out the test. It is only used for in vitro diagnostics for self-testing.

Do not eat, drink or smoke in the area where the specimens or kit are being processed.

Store in a dry place at 2.30T (36.86°F), avoiding places with excess moisture. If foil packaging is damaged or has been opened, please do not use.

This test kit is intended to be used only as a preliminary test and repeated abnormal results should be discussed with a physician or health care professional. Strictly observe the specified time. Use the test only once. Do not disassemble or touch the test window of the test cartridge. The kit should not be frozen or used after the expiry date printed on the packaging. Keep away from children. The used test must be disposed of in accordance with local regulations.

MATERIALS

1. Test cartridge
2. Lancet
3. Capillary dropper
4. A packet of alcohol swabs
5. Diagnostic reagent - Vitamin D Buffer
6. Leaflet and method of use

Materials needed but not provided:

- Timer

PROCEDURE

1. Wash your hands with soap and rinse thoroughly with clean warm water.
2. Leave the box at room temperature before opening it.
3. Open the box, remove the test cartridge and place it on a clean and flat surface. Then remove all other accessories from the box and place them in a convenient place near you. Perform the test within one hour, and you will get the best results if the test is performed

immediately after opening the foil pouch.

4. Carefully pull and discard the released lancet cap.

Use an alcohol swab to clean the fingertip / middle or ring finger a finger/ that will be pierced. Allow to air dry.

5. Press the lancet on the front side where the cap is removed. The nozzle retracts automatically and safely after use. Massage the hand without touching the puncture site, massaging the hand towards the fingertip of the middle or ring finger to be pricked. Holding the hand down, massage the end that is pricked to get a drop of blood.

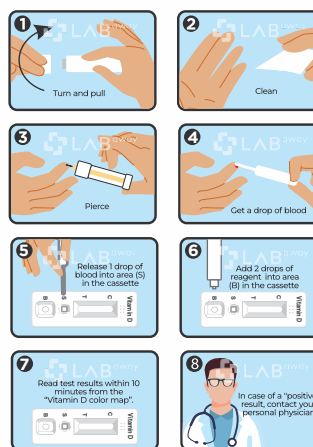
Without squeezing the dropper bulb, place it in contact with the blood. The blood migrates into the capillary dropper through the capillary to the line indicated on the capillary dropper.

6. You can massage your finger again to get more blood if the blood does not reach the indicated line. Avoid air bubbles.

7. Release the blood into the sample well (S) of the cartridge by squeezing the cartridge. Wait for the blood to disperse completely in the well, then clean your finger with alcohol and a swab.

8. Unscrew the cap of the reagent bottle, Vitamin D Buffer, and add 2 drops of solution to the buffer well (B) on the cartridge and start the timer.

9. Wait for the coloured lines to appear. Read the results after 10 minutes. Compare the intensity of the "T" line with the "vitamin D color chart" provided with the kit to get your blood vitamin D level. Do not interpret the result after 20 minutes.



READING THE RESULTS

(Please stick to the illustration and compare the T-Line intensity with the "vitamin D color chart" provided in the kit).

25-OH vitamin D level	Reference values (ng/mL)	
	(ng/mL)	(nmol/l)
Deficit	0-10	0-25
Scarcity	10-30	25-75
Sufficient level	30-100	75-250

Strong shortage

Two clearly coloured lines appear. One is in the control area (C) and the other should be in the smallest area (T). The line intensity in the test area (T) is equal to or darker than the 10 ng/mL line depicted on the color card provided with the kit.

Shortcoming

Two coloured lines appear. One is in the control area (C) and the other should be in the test area (T). The line intensity in the test area (T) is darker than the 30 ng/mL line depicted on the color card provided with the kit and lighter than the 10 ng/mL line depicted on the color card provided with the kit.

In the norm

Two coloured lines appear, one line should always be in the control area (C) and a faint coloured line appears in the test area (T). The intensity of the line in the area (T) shall be equal to or lighter than the 30 ng/mL line depicted on the colour map.

Above the norm

A coloured line appears in the area of the control line (C). No visible coloured line appears in the test line area (T). If the result is excessive, it is advisable to consult a doctor.

Invalid

The control line does not appear. Insufficient sample volume or improper procedural techniques are the most likely causes of control line failure. Review the procedure and repeat the test with a new test. If the problem persists, discontinue use of the test kit immediately and contact your local distributor.

RESTRICTIONS

1. The vitamin D rapid test cartridge provides only a semi-quantitative analytical result. A secondary analytical method must be used to obtain a confirmed result.
2. It is possible that technical or procedural errors as well as other interfering substances in the whole blood sample could lead to erroneous results.
3. As with all diagnostic tests, all results should be considered with other clinical information available to the physician.
4. Other clinically available tests are needed if suspicious results are obtained.

FURTHER INFORMATION

1. How does the vitamin D test work?

In medicine, 25-hydroxy vitamin D is the main storage form of vitamin D in the body. Therefore, the overall status of vitamin D can be determined by detecting the content of 25-hydroxy vitamin D. A level of 25-hydroxy vitamin D less than 30 ng/ml in case of a positive result indicates vitamin D deficiency or insufficiency. Vitamin D supplementation may be recommended in these cases.

2. When should the test be used? The clinical application of 25-hydroxy vitamin D is mainly for the diagnosis, treatment and monitoring of rickets (children), osteomalacia, postmenopausal osteoporosis and renal osteopathy. Vitamin D deficiency is also associated with many other diseases including cancer, cardiovascular disease, autoimmune diseases, diabetes and depression. Monitor your vitamin D levels to determine whether to take vitamin D supplements. The rapid vitamin D test can be used at any time of the day.

3. Could the result be incorrect? The results are accurate as long as the instructions are followed carefully. Nevertheless, the result may be incorrect if the vitamin D rapid test cassette gets wet before performing the test, or if the amount of blood splashed into the sample well is not sufficient, or if the number of buffer drops is less than 2 or more than 3. The capillary dropper provided in the box allows you to make sure that the blood volume collected is correct. Furthermore, due to the immunological principles involved, there are chances of false results in rare cases. It is always recommended to consult a doctor for such tests based on immunological principles.

4. How to interpret the test if the color and intensity of the lines are different?

Please refer to the illustration and compare the intensity of the T line with the "Vitamin D Color Card" provided with the kit.

5. If I read the result after 20 minutes, will the result be reliable? No. The result should be read 10 minutes after adding the buffer. The result is unreliable after 20 minutes.

6. What should I do if the result is insufficient or insufficient? If the result is insufficient or deficient level, it means that the level of vitamin D in the blood is less than 30 ng/ml and you should consult a doctor to show the test result. The doctor will then decide if further analysis should be done.

7. What should I do if the result is sufficient? If the result is sufficient, it means that the vitamin D level is greater than or equal to 30 ng/mL and is within the normal range. A case of vitamin D toxicity (hypercalcemia), although rare, but cancellation is ruled out based on such test results. However, if symptoms persist, it is advisable to consult a physician.

BIBLIOGRAPHY

1. Holick MF March 2006. High incidence of vitamin D inadequacy and health consequences. Mayo Clinic Proceedings. 81(3):353-73.
2. Eriksen EI, Glerup H (2002). Vitamin D deficiency and aging: implications for general health and osteoporosis. Biogerontology. 3 (14): 73-7.
3. Grant WB, Holick M (2005). Vitamin D benefits and needs for optimal health: a review. Alternative Medicine Review.10(2):94-111.

Please check the instructions before using. Not reusable. Store between 4 - 30 °C
DO NOT FREEZE. Do not use after expiry date.

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